MSG No. 59.416 P_Imnn [Type IV, orthorhombic]

Table 1: Wyckoff site: `2a`, site symmetry: `mmm'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,4,6,7,10,11,13,16]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,5,8,9,12,14,15]`

Table 2: Wyckoff site: `2b`, site symmetry: `mmm'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[1,4,6,7,10,11,13,16]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[2,3,5,8,9,12,14,15]`

Table 3: Wyckoff site: `2c`, site symmetry: `mmm'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,4,6,7,10,11,13,16]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[2,3,5,8,9,12,14,15]`

Table 4: Wyckoff site: `2d`, site symmetry: `mmm'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[1,4,6,7,10,11,13,16]`
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[2,3,5,8,9,12,14,15]`

Table 5: Wyckoff site: `4e`, site symmetry: `mm2`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,4,6,7]`
2	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[2,3,5,8]`
3	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[9,12,14,15]`
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[10,11,13,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `mm2`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,4,6,7]`
2	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[2,3,5,8]`
3	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[9,12,14,15]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[10,11,13,16]`

Table 7: Wyckoff site: `4g`, site symmetry: `2'mm'`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[1,7,10,16]`
2	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[2,8,9,15]`
3	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[3,5,12,14]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[4,6,11,13]`

Table 8: Wyckoff site: `4h`, site symmetry: `2'mm'`

No.	position	mapping
1	$\left[x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[1,7,10,16]`
2	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2,8,9,15]`
3	$\left[-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[3,5,12,14]`
4	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[4,6,11,13]`

Table 9: Wyckoff site: `4i`, site symmetry: `m2'm'`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ \frac{3}{4}\right]$	`[1,6,11,16]`
2	$\left[\frac{3}{4},\ -y,\ \frac{1}{4}\right]$	`[2,5,12,15]`
3	$\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[3,8,9,14]`
4	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[4,7,10,13]`

Table 10: Wyckoff site: `4j`, site symmetry: `m2'm'`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ \frac{1}{4}\right]$	`[1,6,11,16]`
2	$\left[\frac{3}{4},\ -y,\ \frac{3}{4}\right]$	`[2,5,12,15]`
3	$\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	`[3,8,9,14]`
4	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	`[4,7,10,13]`

Table 11: Wyckoff site: `8k`, site symmetry: `-1`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[1,5]`
2	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,6]`
3	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[3,7]`
4	$[0,\ 0,\ 0]$	`[4,8]`
5	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[9,13]`
6	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[10,14]`
7	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[11,15]`
8	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[12,16]`

Table 12: Wyckoff site: `8l`, site symmetry: `..m'`

No.	position	mapping
1	$\left[x,\ y,\ \frac{3}{4}\right]$	`[1,16]`
2	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{4}\right]$	`[2,15]`
3	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[3,14]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[4,13]`
5	$\left[-x,\ -y,\ \frac{1}{4}\right]$	`[5,12]`
6	$\left[\frac{1}{2}-x,\ y,\ \frac{3}{4}\right]$	`[6,11]`
7	$\left[x,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[7,10]`
8	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[8,9]`

Table 13: Wyckoff site: `8m`, site symmetry: `m..`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ z\right]$	`[1,6]`
2	$\left[\frac{3}{4},\ -y,\ -z\right]$	`[2,5]`
3	$\left[\frac{3}{4},\ y+\frac{1}{2},\ -z\right]$	`[3,8]`
4	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ z\right]$	`[4,7]`
5	$\left[\frac{3}{4},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,14]`
6	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[10,13]`
7	$\left[\frac{1}{4},\ y,\ \frac{1}{2}-z\right]$	`[11,16]`
8	$\left[\frac{3}{4},\ -y,\ z+\frac{1}{2}\right]$	`[12,15]`

Table 14: Wyckoff site: `8n`, site symmetry: `.m.`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ z\right]$	`[1,7]`
2	$\left[x+\frac{1}{2},\ \frac{3}{4},\ -z\right]$	`[2,8]`
3	$\left[-x,\ \frac{3}{4},\ -z\right]$	`[3,5]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ z\right]$	`[4,6]`
5	$\left[x+\frac{1}{2},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[9,15]`
6	$\left[x,\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[10,16]`
7	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[11,13]`
8	$\left[-x,\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[12,14]`

Table 15: Wyckoff site: `16o`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[x+\frac{1}{2},\ -y,\ -z\right]$	`[2]`
3	$\left[-x,\ y+\frac{1}{2},\ -z\right]$	`[3]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	`[4]`
5	$[-x,\ -y,\ -z]$	`[5]`
6	$\left[\frac{1}{2}-x,\ y,\ z\right]$	`[6]`
7	$\left[x,\ \frac{1}{2}-y,\ z\right]$	`[7]`

continued ...

Table 15

No.	position	mapping
8	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[8]
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[9]
10	$\left[x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[10]
11	$\left[\frac{1}{2}-x,\, y,\, \frac{1}{2}-z\right]$	[11]
12	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[12]
13	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[13]
14	$\left[-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[14]
15	$\left[x+\frac{1}{2},\, -y,\, z+\frac{1}{2}\right]$	[15]
16	$\left[x,\, y,\, \frac{1}{2}-z\right]$	[16]